



TDOT

Department of
Transportation

Tennessee Department of Transportation
Aeronautics Division
7335 Centennial Boulevard
Nashville, Tennessee 37209
615-741-3208

<https://www.tn.gov/tdot/aeronautics.html>

Overview:

The Tennessee Department of Transportation Aeronautics Division (Aeronautics) has developed a Life Cycle Cost Analysis (LCCA) framework that can be used by Aeronautics and its consultants to complete pavement rehabilitation alternatives. LCCA can be a valuable tool for comparing alternatives. The use of a LCCA methodology helps to shift the focus in project selection from the lowest initial cost toward those which have the lowest costs over the life of the pavement cycle cost estimates for TN airport pavement reconstruction projects to assist in the selection of the most cost-effective alternative. LCCA is considered a complete Engineer's Estimate of Probable Costs.

Applied Pavement Technology, Inc. (APTech) has worked with Aeronautics to develop a framework for evaluating pavement reconstruction alternatives to assist with evaluating the most cost-effective alternative. The guidelines provided in the Airfield Asphalt Pavement Technology Program (AAPT) study (AAPT Project 06-06) and Federal Aviation Administration (FAA) guidelines (FAA Advisory Circular 150/5320-6, *Airport Pavement Design and Evaluation*) are used in this release. The developed framework includes future maintenance activities and their costs, intermediate rehabilitation activities, and the generation of a Present Value for the alternatives.

Maintenance policies and timing have been based on analysis of data collected from the Pavement Management Program (PMP) and annual maintenance program. As pavement alternatives are evaluated through the PMP, policies will be revisited. The analysis period has been set based on nationwide data.

Salvage value for the pavement alternatives is based on the assumed remaining value. Residual value, the monetary calculation of the worth of the existing pavement at the end of its life, is added to the value of the existing pavement. Adding value to recycling pavement materials at the end of their life, there is a potential for the suitability of the material for reuse, the future demand for the recycled materials, and the residual value is not used in these calculations.

Additional analysis inputs include discount rate and contingency fees (mobilization and demobilization) established by Aeronautics and should only be revised after discussion with Aeronautics.

If interested in the full report produced as part of developing this LCCA framework, contact the Aeronautics Division.

tics) has sponsored the development of an analytical a reasonable life-cycle cost analysis (LCCA) of aring alternatives of varying costs and service lives. away from alternatives with the lowest initial costs . This workbook assists in the development of life- st with pavement type selection. It is not to be

develop this LCCA framework to compare feasible options. The developed LCCA framework considers n's *Life Cycle Cost Analysis for Airport Pavements* dance in Advisory Circular 150/5320-6G (at the time imework considers construction (or initial) costs, tivities and associated costs, and salvage value to

ted as part of the Tennessee Statewide Pavement nt performance data continues to be collected to 30 years based on reviewed TN PMP and

ining life as a percentage of construction cost. ent at the end of the service life, while certainly are enough uncertainties associated with the erial, and the costs associated with recycling the

lization and engineering). These have been Aeronautics.

etwork, contact Aeronautics.



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Instructions:

This workbook has been set up to automate maintenance cost calculations and cost analyses for Aeronautics' review. This workbook contains privileged information authorized by Aeronautics.

The use of the workbook is fairly straightforward, and the user should be able to obtain a life-cycle cost estimate. Excluding this "Instructions" worksheet and "Overview" worksheet, there are four worksheets: "General Information," "Pay Items," "Maintenance Policies," and "Summary." The "General Information," "Pay Items," and "Maintenance Policies" worksheets are for reference only and should only be altered after the "General Information" worksheet, additional worksheets for anticipated reconstruction.

The following step-by-step procedures are provided to assist in using the workbook. The worksheets fit together. Be sure to follow these directions in the logical, step-by-step order. Use the cut and paste feature as this can write over any functions, commands, or data. When the file is opened so that functions work properly. There have also been multiple monitors are in use. Opening the file in the main monitor should work.

Note: Throughout the workbook, cells that require user-supplied inputs are indicated by a yellow background. Cells that are calculated or auto-filled (and should not be modified by users) are indicated by a light blue background.

1. Go to the "General Information" worksheet and enter in the relevant item information (such as project airport, branch information, pavement area, and so on). After the information is entered, click on the "Calculate" button to complete the cost computations. If any of the items are not applicable (for example, no airport), enter zero.
2. Still in the "General Information" worksheet, click on the "Create Alternatives" button. A separate form will pop up for adding alternatives. The pop-up window includes a list of alternatives. Select which alternatives will be used, update details and description, if desired. Click on "Copy" or "Delete" alternatives. Alternatives can also be reorganized using the "Move" button. When the alternatives are completed, click "Close" and a worksheet for each alternative will be created.
3. After the worksheets have been created for each alternative included in the workbook, enter appropriate pay items in cells C13 through C22 for the alternative (only select pay items that LCCA is intended for paving items only and does not include other items, such as structures). Enter quantities for each selected pay item in cells E13 through E22.
4. Once the pay items and quantities have been filled in, the Net Present Worth (NPW) and costs should be automatically filled in for the alternative. The "Summary" worksheet will show the initial construction cost and NPW for each included alternative.

If there are any issues or questions with the use of the LCCA workbook, please contact the Aeronautics Division.

so as to expedite the overall preparation of life-cycle information and is intended for use only by consultants

le to easily work their way through the development "overview" worksheet, the workbook initially contains " " and "Summary." The Pay Items and Maintenance discussion with Aeronautics. From the General alternatives are created.

orkbook and in understanding how the series of go-by-step fashion to avoid problems. Also, don't use or equations. Make sure to "Enable Content" in Excel on issues with functions running properly when correct function issue, if they occur.

ndicated in grey shading. Most other cells are

is in cells D3 through D17. This is general (and so on) that is needed to identify the project and (example, no shoulders or no paint markings), enter

ives" button at the bottom of the information list. A les the list of available alternatives for this LCCA. ired, and click on "Add." The form can also be used ig the buttons at the bottom of the form. Once d. A "Summary" worksheet is also populated as

he LCCA, go to each worksheet and select the ect the cells you need; leave extra cells blank). This ch as electrical, drainage, and so on. Enter the

orth (NPW) table and summary graph below the pay orksheet will also automatically fill with alternative r worksheet provides a summary table and chart for

use contact Aeronautics.

**Airport Information:**

Airport Name:
Airport ID:
City:
Airport Owner:
State Region:
Design Consultant:
Project Number:
Project Name:
Date of LCCA:

Project Information:

Branch Type:
Branch Name:
Project Type:
Project Description:
Initial Construction Year:
Mainline Area, S.Y.:
Shoulder Area, S.Y.:
Markings, S.F.:
Markings, Type:

LCCA Parameters:

Analysis Period, years:
Discount Rate, percent:
Salvage Value, \$:
Mobilization, percent:
Engineering, percent:
Account Indirect Cost (Yes/No)

Alternative Setup:

View Summary ►

Murfreesboro Municipal Airport	
MBT	
Murfreesboro	
0	
Middle	
Applied Research Associates	
	5935
TDOT Aeronautic Life-Cycle Cost Analysis	
	4/20/2026

Runway	
Runway 1836	
Reconstruction	
Sensitivity Analysis	
	2027
	52777.7
	5433
Reflective	

	30
	3
Remaining service life at end of analysis period (as percent of cost)	
	10
	5
	Yes

Alternative Setup

Division	Division Type	Pay Item No.
Part 3 – Sitework	Default	
	Sitework	P-101-5.1a
	Sitework	P-101-5.1b
	Sitework	P-101-5.2a
	Sitework	P-101-5.2b
	Sitework	P-101-5.2c
	Sitework	P-101-5.2d
	Sitework	P-101-5.3
	Sitework	P-101-5.4a
	Sitework	P-101-5.4b
	Sitework	P-101-5.5a
	Sitework	P-101-5.5b
	Sitework	P-101-5.5c
	Sitework	P-101-5.6
	Sitework	P-152-4.1
	Sitework	P-152-4.2
	Sitework	P-154-5.1
	Sitework	P-154-5.2
	Sitework	P-155-8.1
	Sitework	P-156-8.1
	Sitework	P-157-8.1a
	Sitework	P-157-8.1b
	Sitework	P-158-8.1
Part 4 – Base Courses	Base Course	P-207-5.1
	Base Course	P-208-5.1
	Base Course	P-208-5.2
	Base Course	P-209-5.1
	Base Course	P-209-5.2
	Base Course	P-219-5.1
	Base Course	P-219-5.2
	Base Course	P-220-6.1
Part 5 – Stabilized Base Courses	Stabilized Base Course	P-304-8.1
	Stabilized Base Course	P-306-8.1
	Stabilized Base Course	P-307-7.1
Part 6 – Flexible Pavements	Flexible Pavement	P-401-8.1a
	Flexible Pavement	P-401-8.1b
	Flexible Pavement	P-401-8.1c
	Flexible Pavement	P-403-8.1a
	Flexible Pavement	P-403-8.1b
	Flexible Pavement	P-403-8.1c
	Flexible Pavement	P-407-X.1
Part 7 – Rigid Pavement	Cement Concrete	P-501-8.1a
	Cement Concrete	P-501-8.1b
	Cement Concrete	P-501-8.1c
	Cement Concrete	P-501-8.1d
Part 8 – Surface Treatments	Surface Treatment	P-608-8.1
	Surface Treatment	P-623-8.1
	Surface Treatment	P-626-6.1
	Surface Treatment	P-629-8.1
	Surface Treatment	P-630-7.1a
	Surface Treatment	P-631-7.1b
Part 9 – Miscellaneous	Tack Coat	P-603-5.1
	Joint Sealant	P-605-5.1a
	Marking	P-620-5.1a
	Marking	P-620-5.1b

Grooving
Grooving

P-621-5.1A
P-621-5.1B

Pay Item Description

Pavement Removal, Asphalt
Pavement Removal, Concrete
Crack Repair (Type I), Asphalt
Crack Repair (Type II), Asphalt
Crack Seal, Concrete
Crack Repair, Concrete
Markings Removal
Spall Repair, Asphalt
Full-Depth Repair, Asphalt
Spall Repair, Concrete
Full-Depth Repair, Concrete
Slab Replacement, Concrete
Cold Milling
Unclassified Excavation
Embankment In-Place
Subbase Course
Separation geotextile
Lime Treated subgrade
Cement-Treated Subgrade
Cement Kiln Dust-Treated Subgrade
Lime Kiln Dust-Treated Subgrade
Fly Ash-Treated Subgrade
In-Place Full Depth Reclamation (FDR) Recycled Asphalt Aggregate Base Course
Aggregate Base Course
Separation Geotextile
Crushed Aggregate Base Course
Separation Geotextile
Recycled Concrete Aggregate Base Course
Separation Geotextile
Cement-Treated Soil Base Course
Cement-Treated Base Course
Lean Concrete Base Course
Cement-Treated Permeable Base Course (CTPB)
Asphalt Surface Course
Asphalt Base Course
Asphalt Leveling Course
Asphalt Mixture Surface Course
Asphalt Mixture Base Course
Asphalt Mixture Leveling Course
Asphalt-Treated Permeable Base Course (ATPB)
Concrete Pavement, 5-inch
Concrete Pavement, 6-inch
Concrete Pavement, 9-inch
Concrete Pavement, 11-inch
Emulsified Asphalt Surface Treatment
Emulsified Asphalt Spray Seal Coat
Emulsified Asphalt Slurry Seal Surface Treatment
Thermoplastic Coal Tar Emulsion Surface Treatment
Refined Coal Tar Emulsion Without Additives, Slurry Seal Surface Treatment
Refined Coal Tar Emulsion With Additives, Slurry Seal Surface Treatment
Emulsified Asphalt Tack Coat
Joint Clean and Reseal, Concrete
Markings (prep, markings, reflective media)
Markings (prep, markings, non-reflective media)

Grooving, Asphalt
Grooving, Concrete

Units	Unit Cost	Average Pay Item Unit Cost		
		Middle	West	East
N/A	N/A			
S.Y.	\$5.10			Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
L.F.	\$1.66			
L.F.	\$2.46			
L.F.	\$1.66			
L.F.	\$2.46			
S.F.	\$0.78			
S.F.	\$8.50			Thickness adjustment or mill
S.F.	\$130.00			Assumed Section: 4-inch H
S.F.	\$60.00			Thickness adjustment or mill
S.Y.	\$120.00			Thickness adjustment or mill
S.Y.	\$100.00			
S.Y.	\$13.90			Thickness adjustment or mill
C.Y.	\$8.18			
C.Y.				
C.Y.	\$55.00			
S.Y.				
S.Y.	\$10.00			Thickness adjustment or mill
S.Y.	\$5.00			Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
S.Y.	\$30.00			Thickness adjustment or mill
C.Y.				
S.Y.				
C.Y.	\$77.48			
S.Y.				
C.Y.				
S.Y.	\$15.00			
S.Y.				Thickness adjustment or mill
C.Y.	\$19.00			If S.Y., then thickness adjust
C.Y.				If S.Y., then thickness adjust
C.Y.				If S.Y., then thickness adjust
TON	\$189.40			
TON	\$189.40			
TON	\$189.40			
TON	\$189.40			
TON	\$189.40			
TON	\$189.40			
TON				
S.Y.	\$97.65			Thickness adjustment or mill
S.Y.	\$105.40			Thickness adjustment or mill
S.Y.	\$128.64			Thickness adjustment or mill
S.Y.	\$144.14			Thickness adjustment or mill
S.Y.	\$7.75			
S.Y.	\$7.75			
S.Y.	\$7.75			
S.Y.	\$7.75			
S.Y.	\$7.75			
S.Y.	\$7.75			
GAL	\$5.78			Assume 0.1 gal/SY for mill
L.F.	\$2.58			Assume 12.5x12.5 slabs fo
S.F.	\$1.23			
S.F.	\$0.80			

S.Y.
S.Y.

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MA; 4-inch P-209
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and overlay
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Alternative List

	Name	Type
Alternative 1		HMA-New
Alternative 2		PCC-New

Description	Alt Worksheet Name
	Alt 1 (New HMA)
	Alt 2 (New PCC)



Table 1. New HMA maintenance policies.

Asphalt Maintenance
Maintenance 1
Maintenance 2
Maintenance 3
Maintenance 4
Rehabilitation 1
Maintenance 5
Maintenance 6
Salvage

Table 2. New PCC maintenance policies.

Concrete Maintenance
Maintenance 1
Rehabilitation 1
Salvage

Table 3. First HMA Overlay maintenance po

Asphalt Maintenance
Maintenance 1
Maintenance 2

Maintenance 3
Rehabilitation 2
Maintenance 4
Maintenance 5
Maintenance 6
Salvage

Table 4. PCC rehabilitation maintenance po

Concrete Maintenance
Maintenance 1
Rehabilitation 1
Salvage

Maintenance Item	Rate
Surface treatment, SYD	1.0000
Paint markings, SFT	1.0000
Surface treatment, SYD	1.0000
Paint markings, SFT	1.0000
Patching: 0.75% of area	0.0075
Crack sealing: 3.25% of area as linear feet	0.0325
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Patching: 1.00% of area	0.0100
Crack sealing: 3.50% of area as linear feet	0.0350
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Mill and overlay: 4-inch cold mill, tack coat, and asphalt overlay	1.0000
Pre-overlay crack repair: 1.5% of area as linear feet	0.0150
Pre-overlay patching: 0.5% of area	0.0050
Paint markings: 100% of markings	1.0000
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Patching: 1.25% of area	0.0125
Crack sealing: 4.0% of area as linear feet	0.0400
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Salvage value: 2 years (0.125) of mill and overlay	0.1250

Maintenance Item	Rate
Joint resealing: 100% of joints	1.0000
Crack sealing: 0.25% of area as linear feet	0.0025
Partial-depth patching: 0.13% of area	0.0013
Full-depth patching: 0.30% of area	0.0030
Slab replacements: 2% of area	0.0200
Full-depth patching: 0.30% of area	0.0030
Partial-depth patching: 0.13% of area	0.0013
Crack sealing: 0.30% of area as linear feet	0.0030
Joint sealing: 100% of joints	1.0000
Salvage value: 10 years (0.25) of initial construction	0.2500

licies.

Maintenance Item	Rate
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Patching: 1.25% of area	0.0125
Crack sealing: 4.0% of area as linear feet	0.0400
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000

Surface treatment, SYD	1.0000
Paint markings, SFT	1.0000
Mill and overlay: 4-inch cold mill, tack coat, and asphalt overlay	1.0000
Pre-overlay crack repair: 2% of area as linear feet	0.0150
Pre-overlay patching: 1% of area	0.0050
Paint markings: 100% of markings	1.0000
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Patching: 1.25% of area	0.0125
Crack sealing: 4.0% of area as linear feet	0.0400
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Salvage value: Need for reconstruction (zero salvage)	0.0000

licies.

Maintenance Item	Rate
Joint resealing: 100% of joints	1.0000
Crack sealing: 0.35% of area as linear feet	0.0035
Partial-depth patching: 0.2% of area	0.0020
Full-depth patching: 0.30% of area	0.0030
Slab replacements: 5% of area	0.0500
Full-depth patching: 0.30% of area	0.0030
Partial-depth patching: 0.15% of area	0.0015
Crack sealing: 0.30% of area as linear feet	0.0030
Joint sealing: 100% of joints	1.0000
Salvage value: Need for reconstruction (zero salvage)	0.0000

Year Applied
4
8
12
16
20
24
28
30

Year Applied
19
27
30

Year Applied
4
8

12
16
20
24
28
30

Year Applied
15
25
30

MBT Revenue Data

Revenue Item	Average Daily Reven
100LL Fuel Sales	\$
Jet-A Fuel Sales	\$
Transient Tie-downs	\$
Itinerant Hangar Storage	\$
Tenant Fuel Flowage Fee	-
Business Tenant Revenue	\$
Other Revenue Sources	\$

ue
4,939.57
1,517.81
30.00
10.00
2,344.00
-

Initial Construction Costs:

Pavement Type:
Mainline Pavement Area:
Initial Construction Year:
Asphalt Surface Treatment Type:
Markings Area:

Item No.	Pay Item	
	1 P-152-4.1	Unclassified Excavation
	2 P-209-5.1	Crushed Aggregate Base Course
	3 0	
	4 P-620-5.1a	Markings (prep, markings, reflective m
	5 P-401-8.1a	Asphalt Surface Course
	6 P-603-5.1	Emulsified Asphalt Tack Coat
	7 0	
	8 0	
	9 0	
	10 0	
Subtotal		
	Mobilization	

Total

Net Present Worth
HMA Reconstruction - Alternative 1

Item
Initial Construction
Maintenance 1
Maintenance 1 Indirect Cost
Maintenance 2
Maintenance 2 Indirect Cost
Maintenance 3
Maintenance 3 Indirect Cost
Maintenance 4
Maintenance 4 Indirect Cost

Rehabilitation 1
Rehabilitation 1 Indirect Cost
Maintenance 5
Maintenance 5 Indirect Cost
Maintenance 6
Maintenance 6 Indirect Cost
Salvage
Net Present Worth



New HMA

52777.7
2027

Emulsified Asphalt Surface Treatment
5433

Description

Unclassified Excavation	▼
Crushed Aggregate Base Course	▼
	▼
Markings (prep, markings, reflective media)	▼
Asphalt Surface Course	▼
Emulsified Asphalt Tack Coat	▼
	▼
	▼
	▼
	▼

Year Applied
2027
4
4
8
8
12
12
16
16

20
20
24
24
28
28
30

Alternative :



S.Y.

Select alternative treatment, if planned.
S.F.

Unit

- C.Y.
- C.Y.
- N/A
- S.F.
- TON
- GAL
- N/A
- N/A
- N/A
- N/A

Actual Cost	
	\$5,306,577
	\$482,830
	\$53,048
	\$482,830
	\$53,048
	\$547,129
	\$70,731
	\$561,004
	\$70,731

Average Daily Revenue:	\$ 8,841.38
Maintenance 1 Duration (Days)	6
Maintenance 2 Duration (Days)	6
Maintenance 3 Duration (Days)	8
Maintenance 4 Duration (Days)	8
Rehabilitation 1 Duration (Days)	15
Maintenance 5 Duration (Days)	6
Maintenance 6 Duration (Days)	8

Qty		Unit Cost	Item Cost	
	30,787	\$8.18	\$251,796	-
	23,457	\$77.48	\$1,817,478	-
		N/A	\$0	-
	5,433	\$1.23	\$6,682	-
	14,349	\$189.40	\$2,717,697	
	5,278	\$5.78	\$30,508	
		N/A	\$0	
		N/A	\$0	
		N/A	\$0	
		N/A	\$0	
			\$4,824,160	
			\$482,416	
			\$5,306,577	

Discounted Cost
\$5,306,577
\$428,988
\$47,133
\$381,150
\$41,877
\$383,745
\$49,609
\$349,599
\$44,077

Initial Construction Costs:

Pavement Type:	New PCC	
Mainline Pavement Area:		52777.7
Initial Construction Year:		2027
Markings Area:		5433

Item No.	Pay Item	Description
1	P-152-4.1	Unclassified Excavation
2	P-501-8.1c	Concrete Pavement, 9-inch
3	P-209-5.1	Crushed Aggregate Base Course
4	P-620-5.1a	Markings (prep, markings, reflective media)
5	0	
6	0	
7	0	
8	0	
9	0	
10	0	

Subtotal
Mobilization

Total

**Net Present Worth
PCC Reconstruction - Alternative 2**

Item	Year Applied
Initial Construction	2027
Maintenance 1	19
Maintenance 1 Indirect Cost	19
Rehabilitation 1	27
Rehabilitation 1	27
Salvage	30
Net Present Worth	

Alternative 2



Average Daily Revenue:	\$	8,841.38
Maintenance 1 Duration (Days)		9
Rehabilitation 1 Duration (Days)		18

S.Y.

S.F.

Unit	Qty	Unit Cost
C.Y.	20,525	\$8.18
S.Y.	52,778	\$128.64
C.Y.	8,796	\$77.48
S.F.	5,433	\$1.23
N/A		N/A
N/A		N/A
N/A		N/A
N/A		N/A
N/A		N/A
N/A		N/A

Actual Cost	Discounted Cost
\$8,410,245	\$8,410,245
\$306,564	\$174,829
\$79,572	\$45,379
\$428,406	\$192,864
\$159,145	\$71,645
-\$2,102,561	-\$866,227
	\$8,028,734

Hidden

550
555.5556
0.99

Item Cost

\$167,864
\$6,789,577
\$681,554
\$6,682
\$0
\$0
\$0
\$0
\$0
\$0
\$7,645,677
\$764,568

\$8,410,245

-

Hidden

General InformationInstructions

Alternative	Name	Initial Construction	Present Worth	Alternative Description
HMA	Alternative 1	\$ 5,306,576.53	\$ 8,809,266.42	5" P401 on 16" P209
PCC	Alternative 2	\$ 8,410,244.97	\$ 8,028,734.49	8" P501 on 6" P209

LCCA SUMMARY

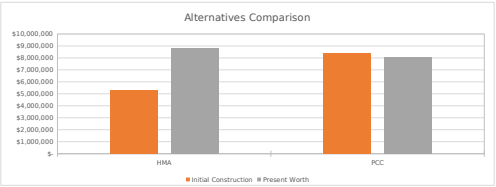
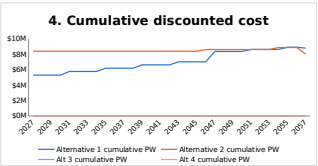
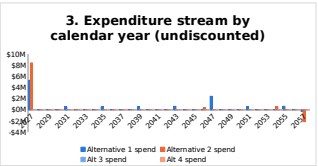
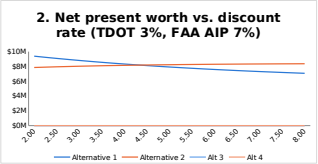
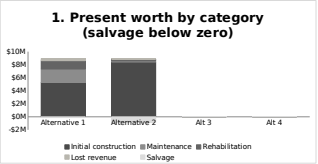
Everything from this column onward calculates from the alternative worksheets and updates by itself. Columns A-E and "Chart 1" are written by the Alternative Setup form as below.

Worksheet	Alternative	Type	Initial construction	Maintenance PW	Rehabilitation PW	Lost revenue PW	Salvage PW	Net present worth	vs. lowest NPW	Closure days in period	Runway availability
Ait 1 (New HMA)	Alternative 1	HMA-New	\$5,306,577	\$2,033,261	\$1,274,870	\$333,136	(\$118,979)	\$8,809,266	\$790,532	57	99.48%
Ait 2 (New PCC)	Alternative 2	PCC-New	\$8,410,245	\$174,829	\$264,509	\$45,379	(\$866,227)	\$8,028,734	-	27	99.75%

Lowest present worth: Alternative 2 | margin to next: \$789,532 | 3% over 30 years | lost revenue: Yes

At the FAA AIP rate of 7% the lowest-cost alternative changes: see chart 2.

CHARTS



How to read: 1 shows where each alternative's cost sits; 2 whether the lowest-cost alternative holds at other discount rates (FAA AIP uses 7%); 3 and 4 when the money is spent and when the higher first cost is paid back; 5 how often and how long the runway closes.